1) Project Name

FoodWise: AI-Powered Expiry-Aware Recipe Recommendation for Sustainable Food Consumption (SDG 12.3)

2) Overview

Household and institutional food waste represents an escalating environmental and economic crisis. The United Nations Environment Programme (UNEP) reports that over 1.05 billion tonnes of food were wasted globally in 2022, with 60% occurring directly at the household level and 28% in food-service operations. A major contributing factor to this waste is the inadvertent spoilage of perishable ingredients when households, soup kitchens, or community pantries lack culinary ideas to utilize items already in stock before their use-by dates.

FoodWise addresses this challenge by designing an intelligent, inventory-aware recipe recommendation system. Users enter their available ingredients, stock quantities, and use-by dates. FoodWise normalizes these inputs, cross-references an indexed recipe corpus, and produces ranked recipe suggestions that prioritize items nearing expiration while transparently highlighting any missing ingredients. Through proactive stock matching and expiry prioritization, FoodWise directly advances UN Sustainable Development Goal 12 (Responsible Consumption and Production), specifically Target 12.3 (halving per capita global food waste by 2030), while indirectly supporting SDG 2 (Zero Hunger) through maximized nutritional resource efficiency.

3) Background

Traditional culinary search engines and recipe platforms rely almost exclusively on user queries focused on recipe titles or popularity metrics, failing to account for real-time pantry inventory states or the critical temporal dimension of food perishability. When cooks purchase or receive food without an immediate recipe in mind, high-spoilage items (such as leafy greens, fresh dairy, and proteins) frequently expire unused. Small-scale soup kitchens and community food centers face similar operational bottlenecks, receiving diverse bulk donations that must be quickly processed before spoilage.

Natural Language Processing (NLP) and Information Retrieval (IR) methodologies provide an effective foundation to solve this problem. Recipes can be structured as sets and bag-of-words vectors over normalized ingredient vocabularies. Our Exploratory Data Analysis (EDA) reveals that common staple items (e.g., salt, butter, sugar, onions, olive oil) appear in 20% to 47% of recipes. Consequently, naive set overlap trivially favors recipes loaded with ubiquitous pantry staples rather than recipes that solve perishable ingredient bottlenecks. By training an Inverse Document Frequency (TF-IDF) representation, the system learns to automatically down-weight common staples and assign high retrieval weight to rare, distinctive perishable ingredients. Furthermore, latent semantic embeddings (such as TruncatedSVD and dense transformer representations) enable culinary synonym handling and flexible ingredient substitution, allowing an expiry-prioritized scoring heuristic to dynamically balance semantic suitability, ingredient coverage, and use-by urgency.

4) Key Objectives / Business Objectives

a) Research Questions:

1. **Distributional Suitability:** What are the structural and distributional characteristics of recipe datasets, and how do ingredient diversity and category balance impact pantry-driven recipe retrieval?
2. **Feature Weighting:** How effectively can TF-IDF inverse document frequency weighting mitigate the over-representation of ubiquitous staple ingredients (e.g., salt, butter, onion) compared to unweighted Jaccard overlap?
3. **Expiry Prioritization:** How can user-entered use-by dates be integrated into multi-objective ranking functions to elevate recipes that utilize urgent stock without concealing missing items?
4. **Data Quality Bottlenecks:** What structural limitations exist in crowdsourced recipe data (e.g., ingredient-to-quantity array length mismatches, nested nulls, non-standardized units), and why must initial matching rely on normalized ingredient presence rather than uncurated mass constraints?
5. **Inventory Coverage Feasibility:** Does an illustrative real-world pantry inventory achieve sufficient ingredient coverage across an 8,000-recipe corpus to surface viable, complete meal suggestions?
6. **Semantic Generalization:** Can dense semantic representations capture culinary ingredient substitutions and synonyms more effectively than exact lexical token matching?
7. **Ethical and Safety Boundaries:** How can system design ensure that recipe suggestions do not encourage unsafe consumption, given that software cannot biologically infer food safety from entered dates alone?

b) Key Steps:

1. **Curate Reproducible Dataset:** Filter and extract a reproducible working sample of 8,000 recipes from the Food.com corpus using seed 42 reservoir sampling, enforcing rules on valid titles, 3–20 distinct ingredients, and positive known servings.
2. **Conduct Comprehensive EDA:** Perform in-depth exploratory data analysis covering missing value audits, array-level quality checks, univariate/bivariate distributions, ingredient frequencies, and co-occurrence patterns.
3. **Evaluate Inventory Feasibility:** Conduct an empirical sustainability-oriented coverage feasibility study by evaluating an illustrative stock pantry against all 8,000 recipes.
4. **Develop NLP Retrieval Models:** Construct normalized ingredient tokenizers, train TF-IDF vectorizers, and build latent semantic representations (LSA TruncatedSVD) to represent recipe ingredient vectors.
5. **Formulate Expiry-Aware Ranking:** Design and implement a multi-objective composite ranking function balancing TF-IDF cosine similarity, ingredient coverage, and urgent item utilization.
6. **Implement Data Integrity Validation:** Establish automated functional integrity assertions verifying sample invariants, schema compliance, and output artifact completeness.
7. **Integrate External Nutritional Standards:** Establish linkage pathways to USDA FoodData Central to validate portion scaling and macronutrient profiles for future phases.
8. **Deploy Prototype Interface & Deliverables:** Package the codebase, documentation, and executed notebooks for open-source dissemination and prepare in-class presentation materials.

5) Methods and Workflow

a) Datasets:

- **Primary Corpus (Food.com Recipes and Reviews):** An open-source curated working sample of exactly 8,000 real recipes (seed 42) retaining 11 core attributes: recipe identifier (recipe_id), title (name), culinary category (category), normalized ingredients (ingredients), raw source ingredients (source_ingredients), raw source quantities (source_quantities), serving yield (servings), instruction steps (instructions), calories (calories_source), protein (protein_source), and publication timestamp (published_at). Source publisher states CC0: Public Domain.
- **Illustrative Stock Inventory:** A structured JSON inventory (data/example_inventory.json) representing realistic pantry stock with normalized ingredient names, recorded use-by dates, usability flags, and recorded quantities, evaluated as of a fixed reference date.
- **Supplementary Reference Benchmark:** USDA FoodData Central (Agricultural Research Service) identified for secondary nutritional validation, ingredient synonym mapping, and standardized portion-weight matching in subsequent development phases.

b) Data Cleaning and Preprocessing:

1. **Array Parsing without Execution:** Parse serialized string list representations (e.g., R-style vector syntax) using safe regular expressions, extracting tokens while handling nested missing entries (None/NA) safely without executing code.
2. **Ingredient Text Normalization:** Apply NFKC Unicode normalization, lowercase conversion, punctuation stripping, and whitespace collation to create uniform ingredient keys across recipes and inventory inputs.
3. **Sampling and Quality Filtering:** Enforce strict filtering criteria: named recipes only, positive known servings, and 3 to 20 distinct normalized ingredients. Filter duplicate ingredient sets to prevent identical recommendations before seed-42 sampling.
4. **Data Quality & Integrity Auditing:** Audit top-level completeness (100% complete, 0 missing values across all 11 fields; 100% unique recipe_id values). Uncover critical array-level discrepancies: 77.6% (6,209 recipes) exhibit length mismatches between ingredients and quantities, and 36.2% (2,897 recipes) have missing quantity vector entries. Consequently, restrict retrieval matching to normalized ingredient presence rather than unvalidated mass constraints.
5. **Feature Vectorization Pipeline:** Extract distinct normalized ingredient vocabularies (over 3,400 unique ingredients) and construct token matrices for retrieval modeling.

c) Modelling:

1. **Lexical Baseline (Set Overlap / Jaccard):** Measure set-theoretic intersection and Jaccard similarity between available pantry items and recipe ingredient sets to provide an intuitive, interpretable baseline.
2. **TF-IDF Vector Space Model:** Fit a Term Frequency-Inverse Document Frequency vectorizer over the corpus to learn inverse document frequencies, heavily weighting distinctive ingredients while penalizing ubiquitous staples. Recipes are scored via cosine similarity against the pantry query vector.
3. **Dense Semantic Representation (LSA & Embeddings):** Project the TF-IDF matrix into 64 latent semantic dimensions using TruncatedSVD to capture ingredient co-occurrences. Evaluate pre-trained

sentence transformer embeddings for future semantic synonym resolution and multilingual Turkish-to-English translation.

4. **Expiry-Prioritized Composite Scoring Function:** Formulate a multi-factor ranking algorithm combining semantic relevance, inventory coverage, and date urgency: Score = 0.60 * CosineSimilarity + 0.30 * IngredientCoverage + 0.10 * UrgentCoverage. Filter out entered unusable or expired stock at the boundary; rank recipes with 100% available ingredient types first, followed by ranked partial matches displaying missing items.

d) Deliverables:

1. **Curated Reproducible Sample:** 8,000-record JSONL dataset (data/recipes_sample.jsonl) with complete metadata and extraction scripts (scripts/prepare_data.py).
2. **Executed Exploratory Data Analysis Notebook:** Fully executed, verified Jupyter notebook (notebooks/FoodWise_SDG12_EDA.ipynb) containing descriptive statistics, data quality audits, distribution plots, ingredient co-occurrence tables, and sustainability feasibility analyses.
3. **Data Quality and Coverage Reports:** Generated summary CSV artifacts (missing_values_report.csv, inventory_ingredient_coverage.csv) and 12 high-resolution visualization charts in outputs/figures/.
4. **Structured Project Concept Note & Report:** Comprehensive project proposal documents in both Microsoft Word (.docx) and PDF formats aligned with the course directive and SDG 12.3 framework.
5. **Modular Python Retrieval Codebase:** Production-grade source modules (src/foodwise.py, src/analysis.py) implementing parsers, TF-IDF vectorizers, LSA models, and ranking heuristics.
6. **Presentation & Dissemination Package:** Slide deck materials and open-source project repository documentation for in-class presentation and future development phases.

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